DEVELOPMENT OF REAL TIME UNDERGROUND MONITORING SYSTEM USING UNMANNED VEHICLE

A PROJECT REPORT

Submitted by

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in

INFORMATION TECHNOLOGY

KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY

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5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
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11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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BONAFIDE CERTIFICATE

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Submitted for the project viva-voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

The unmanned vehicle in this project leverages advanced IoT and Raspberry Pi technologies to enable comprehensive monitoring of cave environments, providing valuable data for research in fields like geology, biology, hydrology, and cultural anthropology. The vehicle is equipped with a suite of sensors that measure various environmental parameters, including air quality, temperature, humidity, lighting, soil condition, and object detection. These sensors gather critical data, which is then processed by AI algorithms to identify anomalies, hazards, and other important environmental changes. The system also incorporates GPS technology, enabling real-time data transmission and remote monitoring by researchers, regardless of the cave's location or accessibility. In addition to monitoring environmental conditions, the unmanned vehicle is designed to autonomously navigate and detect hazards such as poor air quality, extreme darkness, water flow, and potential geological threats like collapses, flooding, or landslides. When such dangers are identified, the vehicle is programmed to retreat to a safer area to avoid damage. This proactive approach significantly enhances the safety of cave exploration, reducing the risks to human researchers and improving the overall efficiency of data collection. By utilizing this system, researchers will have a more reliable and effective tool for studying remote and hazardous cave environments, ensuring a safer and more informed exploration process.

ACKNOWLEDGEMENT

We wish to express our sincere thanks to our beloved, respectful and honorable Chairman **Dr. PSK. R. PERIASWAMY** for providing immense facilities in our institution.

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We proudly render our immense gratitude to our Head of the Department of Information Technology **Dr.K.MUTHUMANICKAM, M.E., Ph.D.,** for his effective leadership, encouragement and guidance in the project.

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LIST OF ABBREVIATIONS

IoT	**I**nternet **o**f **T**hings
AI	**A**rtificial **I**ntelligence
FOT	**F**iber **O**ptic **T**echnology
WSN	**W**ireless **S**ensor **N**etworks
YOLO	**Y**ou **O**nly **L**ook **O**nce
DHT	**D**igital **H**umidity and **T**emperature Sensor
Wi-Fi	**W**ireless **F**idelity
GPS	**G**lobal **P**ositioning **S**ystem

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

The The exploration of cave environments is essential for advancing research across multiple scientific disciplines, including geology, biology, hydrology, and cultural anthropology. However, these environments pose considerable challenges to researchers due to poor visibility, unstable rock formations, and hazardous air quality. These challenges can hinder effective exploration and data collection, making cave exploration risky and labour-intensive. To address these issues, this project focuses on developing an unmanned vehicle powered by Internet of Things (IoT) and Raspberry Pi technologies to enhance the safety, efficiency, and effectiveness of underground exploration. The system will offer real-time environmental monitoring, hazard detection, and data collection, ensuring the safety of researchers and providing valuable data for scientific studies.

The core objective of this project is to design and implement an unmanned vehicle using advanced technologies such as Fiber Optic Technology (FOT), Artificial Intelligence (AI) algorithms, and GPS tracking. The vehicle will be equipped with a range of sensors to measure essential environmental parameters such as air quality, temperature, humidity, soil conditions, and the presence of toxic gases. These measurements will be transmitted in real-time via GPS to a remote monitoring system, allowing researchers to receive up-to-date data without needing to be physically present in the hazardous environment. The vehicle will also feature autonomous navigation capabilities, enabling it to move through caves while avoiding obstacles like rocks and water. Additionally, it will be programmed to detect and respond to potential risks such as cave collapses, flooding, and low oxygen levels, crucial for ensuring safe and effective exploration. The system consists of several integrated modules.

The Environmental Sensing and Data Collection module incorporates various sensors to gather real-time environmental data, while AI algorithms process this data to detect hazards and environmental anomalies.The Autonomous Navigation and Obstacle Avoidance module uses depth sensors and AI-driven path planning to help the vehicle navigate safely through challenging cave terrains. The GPS Tracking module ensures precise location monitoring and continuous communication with the remote monitoring system, ensuring safe navigation and real-time updates. The system's ability to provide real-time hazard detection and data analysis offers new possibilities for safer and more effective exploration of complex and dangerous environments. By integrating IoT, AI, and Raspberry Pi technologies, this project offers a transformative solution for cave exploration. The system's ability to provide real-time hazard detection and data analysis opens new possibilities for scientific research in underground environments. Ultimately, this unmanned vehicle will serve as a valuable tool for researchers, reducing risks while improving the quality and efficiency of cave exploration.

1.2 OVERVIEW

Cave exploration plays a crucial role in advancing research across multiple scientific fields, including geology, biology, and anthropology. These environments, however, pose significant challenges due to factors such as poor visibility, unstable rock structures, and hazardous air quality. Navigating caves can be dangerous for researchers, and traditional methods often require significant human involvement in potentially risky environments. To address these issues, this project aims to develop an unmanned vehicle equipped with cutting-edge Internet of Things (IoT) technology, AI algorithms, and GPS tracking for real-time underground monitoring.

The main goal is to enhance both safety and efficiency during exploration while collecting valuable environmental data to support scientific studies. The unmanned vehicle will be equipped with a variety of sensors designed to measure critical environmental parameters, such as air quality, temperature, humidity, and the presence

of toxic gases. These sensors will continuously monitor cave conditions and collect real-time data. The data will be transmitted via GPS to a remote monitoring system, which will allow researchers to analyse cave environments from a safe distance. It will be programmed to detect and avoid obstacles such as rocks, narrow pathways, and water bodies. If it encounters hazardous conditions such as cave collapses or flooding, the vehicle will intelligently adjust its path or retreat to a safer area. This dynamic adaptation enhances the reliability and safety of the exploration process, allowing researchers to explore deeper and more dangerous cave sections without direct intervention. This real-time data transmission reduces the need for human presence in potentially hazardous cave areas, minimizing exposure to risks like low oxygen levels or hazardous gases. Additionally, the vehicle will incorporate AI-driven autonomous navigation, which will enable it to move safely through the complex cave system. The vehicle will be able to navigate around obstacles like rocks, narrow pathways, and water flows. If it detects hazardous conditions such as cave collapses or flooding, the vehicle will automatically adjust its path or retreat to avoid danger, ensuring the safety.

Additionally, the system will include a GPS tracking and communication module, which will provide continuous location monitoring. This ensures that the vehicle's movements can be tracked in real time, allowing researchers to maintain control over the exploration process. The integration of AI and IoT technology not only improves safety but also enhances data collection efficiency, making it possible to conduct long-term studies with minimal human intervention. By leveraging these advanced technologies, the proposed unmanned vehicle provides a revolutionary approach to cave exploration. It enhances safety, streamlines data collection, and enables more effective scientific research in extreme underground environments. The unmanned vehicle will be equipped with an array of sensors to measure essential environmental parameters, including air quality, temperature, humidity, and the presence of toxic gases. These sensors will continuously monitor cave conditions, and the collected data will be transmitted in real time via GPS to a remote monitoring system.

CHAPTER 2

LITERATURE REVIEW

1. Manjunath, A. S., Prakash, K. R., & Shobha, M. P. (2021). "Vehicle Number Plate Detection and Recognition Using YOLO-V3 and OCR." IEEE Transactions on Image Processing, 58(9), 1874–1890.

This study focuses on improving Automated Number Plate Recognition (ANPR) by combining the YOLO-V3 deep learning model with Optical Character Recognition (OCR) techniques. The proposed approach enhances plate detection accuracy under challenging conditions such as motion blur, low lighting, and distorted number plates. A key contribution of this research is the integration of adaptive filtering techniques to remove background noise, which significantly improves character segmentation. The YOLO-V3 model is optimized for real-time performance, making it suitable for embedded systems and IoT-based smart city applications. The scalability of the system allows seamless integration with cloud-based databases, facilitating efficient data handling and security.

2. Sivakoti, K., & Rao, P. (2023). "Toll Collection Automation: Vehicle Detection and Classification Using YOLO and Ensemble OCR." IEEE Transactions on Cybernetics, 47(3), 785–799.

This research presents an automated toll collection system that leverages deep learning-based vehicle detection and classification models. The system employs YOLO for vehicle localization and an ensemble of OCR models to enhance number plate recognition accuracy across different font styles, angles, and lighting conditions. By implementing deep feature extraction and adaptive preprocessing techniques, the system effectively handles occlusions, motion blur, and partial plate visibility. A critical advantage of this model is its ability to operate in real-time toll booth environments, significantly reducing transaction processing times and minimizing human intervention. Additionally, the study evaluates the impact of edge computing and cloud-based storage in improving efficiency and scalability. Field tests conducted on various highways confirm the robustness of the approach.

3. Mathew, T. R., & Paul, P. (2022). "Machine Learning-Based Automated Vehicle Identification Using OpenCV and Deep Learning." IEEE Transactions on Artificial Intelligence, 22(4), 955–972.

This paper explores a hybrid machine learning approach for automated vehicle identification, integrating OpenCV-based image processing with deep learning models. Unlike conventional rule-based methods, which often struggle with variations in plate designs and environmental conditions, this system dynamically improves its accuracy through continuous learning. The model is trained on a diverse dataset covering multiple vehicle types, angles, and weather conditions, ensuring robust performance. By leveraging GPU-accelerated computation, the system achieves high-speed processing, making it viable for real-time applications such as traffic surveillance, automated toll booths, and smart parking solutions. The study also proposes an anomaly detection mechanism that identifies fraudulent license plates, ensuring higher security in automated vehicle tracking systems.

4. Nguyen, H. T., & Tran, D. M. (2020). "Deep Learning-Based Traffic Surveillance System for Smart Cities." Journal of Intelligent Transportation Systems, 31(5), 623–638.

This research introduces a deep learning-powered traffic surveillance system that integrates YOLO and convolutional neural networks (CNNs) to detect and classify vehicles in real time. The system is designed for large-scale smart city applications, offering functionalities such as traffic density estimation, vehicle speed monitoring, and automatic violation detection (e.g., illegal parking, red-light violations). A cloud-based infrastructure enables seamless data storage, allowing city authorities to monitor traffic conditions remotely and respond to incidents proactively. The proposed model achieves high detection accuracy across varying lighting conditions, weather conditions, and road types. Comparative evaluations against traditional traffic surveillance techniques indicate that this system offers improved efficiency, reduced computational costs, and enhanced adaptability.

5.Chen, L., & Wang, Y. (2021). "License Plate Recognition in Adverse Weather Conditions Using Hybrid CNN-RNN Models." IEEE Transactions on Intelligent Transportation Systems, 29(2), 432–448.

This paper proposes a hybrid deep learning framework that combines convolutional neural networks (CNNs) and recurrent neural networks (RNNs) for robust license plate recognition under challenging weather conditions such as fog, heavy rain, and low lighting. The CNN component enhances feature extraction, improving plate detection performance, while the RNN component processes sequential character recognition with high precision. The researchers implement data augmentation strategies, including synthetic noise addition and contrast enhancement, to make the model more resilient against environmental distortions. Performance evaluations indicate that the proposed system significantly outperforms conventional CNN-based models, achieving an average accuracy of 96% in real-world scenarios. Furthermore, the study discusses potential applications in smart parking, automated toll booths, and law enforcement surveillance systems.

6. Patel, S., & Desai, R. (2023). "Automated Parking Management System Using YOLO-V5 and RFID Integration." International Journal of Computer Vision, 45(6), 1102–1118.

This study presents an advanced parking management system that integrates YOLO-V5-based vehicle detection with RFID technology for secure and automated access control. The system enhances security by linking license plate recognition with RFID-based authentication, reducing unauthorized vehicle entry. A key feature of this approach is its adaptability to different parking environments, including indoor and outdoor spaces with varied lighting conditions. The researchers trained the model on a diverse dataset, including vehicles from multiple countries and different plate formats, to improve recognition accuracy. Real-world implementations in corporate parking lots and public parking garages demonstrated a 98% success rate in vehicle identification and a substantial reduction in manual interventions.

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CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Cave exploration has traditionally relied on manual methods, where researchers and explorers enter underground environments to study geological formations, biodiversity, and environmental conditions. These conventional approaches pose significant risks due to unstable cave structures, low visibility, and exposure to toxic gases. The existing systems for underground monitoring typically involve handheld sensors, fixed monitoring stations, and human-operated robotic systems, which have several limitations.

Current underground monitoring solutions also suffer from delayed data transmission, as many do not support real-time communication. This delay can be critical in emergency situations where immediate hazard detection is required. Additionally, existing systems often lack integrated AI-based decision-making, reducing their capability to autonomously adapt to changing environmental conditions. By leveraging IoT, AI, and autonomous navigation, the proposed unmanned vehicle will overcome these challenges, offering real-time monitoring, intelligent hazard detection, and seamless data transmission. This will significantly improve safety and efficiency in cave exploration and environmental research.

3.2 DRAWBACKS

- High Initial Cost
- Technical Complexity
- Environmental Interference
- Limited Range and Coverage
- Vulnerability to Obstructions or Damage
- Regulatory and Legal Challenges

3.3 PROPOSED SYSTEM

Cave exploration has traditionally relied on manual methods, where researchers and explorers enter underground environments to study geological formations, biodiversity, and environmental conditions. These conventional approaches pose significant risks due to unstable cave structures, low visibility, and exposure to toxic gases. The existing systems for underground monitoring typically involve handheld sensors, fixed monitoring stations, and human-operated robotic systems, which have several limitations. Human-operated robotic systems offer some level of automation; however, they require constant remote control, making them inefficient in complex and hazardous environments.Current underground monitoring solutions also suffer from delayed data transmission, as many do not support real-time communication. This delay can be critical in emergency situations where immediate hazard detection is required.

Additionally, existing systems often lack integrated AI-based decision-making, reducing their capability to autonomously adapt to changing environmental conditions. The limitations of these systems highlight the need for an advanced solution. By leveraging IoT, AI, and autonomous navigation, the proposed unmanned vehicle will overcome these challenges, offering real-time monitoring, intelligent hazard detection, and seamless data transmission.

The proposed system introduces an advanced unmanned vehicle equipped with IoT, AI algorithms, and GPS tracking for real-time underground monitoring. Unlike traditional methods, this system will operate autonomously, reducing human exposure to hazardous environments while improving data collection efficiency.AI-driven autonomous navigation will enable the vehicle to move safely through caves while detecting and avoiding obstacles such as rocks, water, and unstable surfaces. The integration of AI-based decision-making allows the vehicle to assess environmental risks and adjust its path accordingly.

IoT-based communication will allow real-time data transmission, ensuring that researchers receive timely alerts on changing cave conditions. A web-based dashboard will display live sensor readings, vehicle status, and hazard notifications. Historical data will also be stored for further analysis, allowing researchers to study long-term environmental changes and trends.

Additionally, the system will feature AI-powered image processing techniques such as YOLO (You Only Look Once) for real-time object detection and a dehaze algorithm to enhance visibility in low-light or foggy conditions. These technologies will ensure that the vehicle can capture clear images and detect hazards accurately, improving the quality of data collected.

By integrating advanced technologies, the proposed system will revolutionize underground exploration. It will improve safety by minimizing human risk, enhance efficiency through automation, and contribute significantly to environmental research. This unmanned vehicle will set a new standard in cave monitoring, with potential applications extending to mining, underwater tunnels, and disaster-prone areas.

3.4 ADVANTAGES

- Accurate Environmental Monitoring
- Autonomous Hazard Detection and Avoidance
- Enhanced Safety Mechanism
- Advanced Object Detection
- Continuous GPS Tracking
- Integrated Data Visualization
- Reliable Data Transmission

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

- Microcontroller : 8051
- Ultrasonic Sensor : HC-SR04
- Air Quality Sensor : MQ2, MQ8, MQ135
- Temperature and Humidity Sensor : DHT11
- Camera : ESP32 Camera
- Soil Condition Sensor : Capacitance Sensor
- GPS Module : Real-time location tracking
- Processor : 64GB RAM

4.2 SOFTWARE REQUIREMENTS

- Server Side : Python 3.9+ (64-bit)
- Client Side : HTML, CSS, Bootstrap
- IDE : VS Code
- Back end : MySQL 8.0
- Server : Flask / FastAPI (for web interface)
- OS : Windows 11

CHAPTER 5

SYSTEM DESIGN

5.1 PROPOSED ARCHITECTURE

The proposed system aims to develop an autonomous, unmanned vehicle designed specifically for cave exploration, utilizing advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), and Raspberry Pi-based computing. This system will enhance data collection efficiency while ensuring the safety of researchers by eliminating the need for direct human presence in hazardous cave environments.

The autonomous vehicle will be equipped with various environmental sensors to monitor key parameters, including air quality, temperature, humidity, soil conditions, lighting, and potential obstacles. AI-driven image processing will enhance real-time imaging capabilities, allowing the system to adapt to its caves surroundings dynamically and improve hazard detection. GPS and wireless communication technologies will enable real-time data transmission, providing researchers with up- to-date insights on cave conditions.

Additionally, the system will incorporate machine learning algorithms that allow it to learn from its environment, improving its navigation and obstacle avoidance over time.By leveraging automation and remote monitoring capabilities, this project aims to create a cost-effective, reliable, and efficient solution for cave exploration. The proposed system not only enhances scientific research but also minimizes the risks associated with traditional exploration methods, making cave studies more accessible and data-driven.

The system's user interface is designed to provide a simple and effective way for operators to monitor and control the vehicle.Operators can use the dashboard to adjust system settings, review historical data, and issue remote commands. The user interface also includes graphical representations of the vehicle’s path and environmental data, making it easier for operators to interpret information and make informed decisions.

Another significant advantage of the proposed system is its scalability and adaptability. The AI-based learning models allow the system to adapt to different underground environments and improve its performance over time. The machine learning algorithms analyze historical data and environmental patterns to refine the system's navigation and hazard detection capabilities. This allows the system to operate effectively in diverse underground settings, including caves, tunnels, mines, and disaster zones. The ability to adapt to different environments and improve performance through learning makes the proposed system highly versatile and valuable for a wide range of applications.

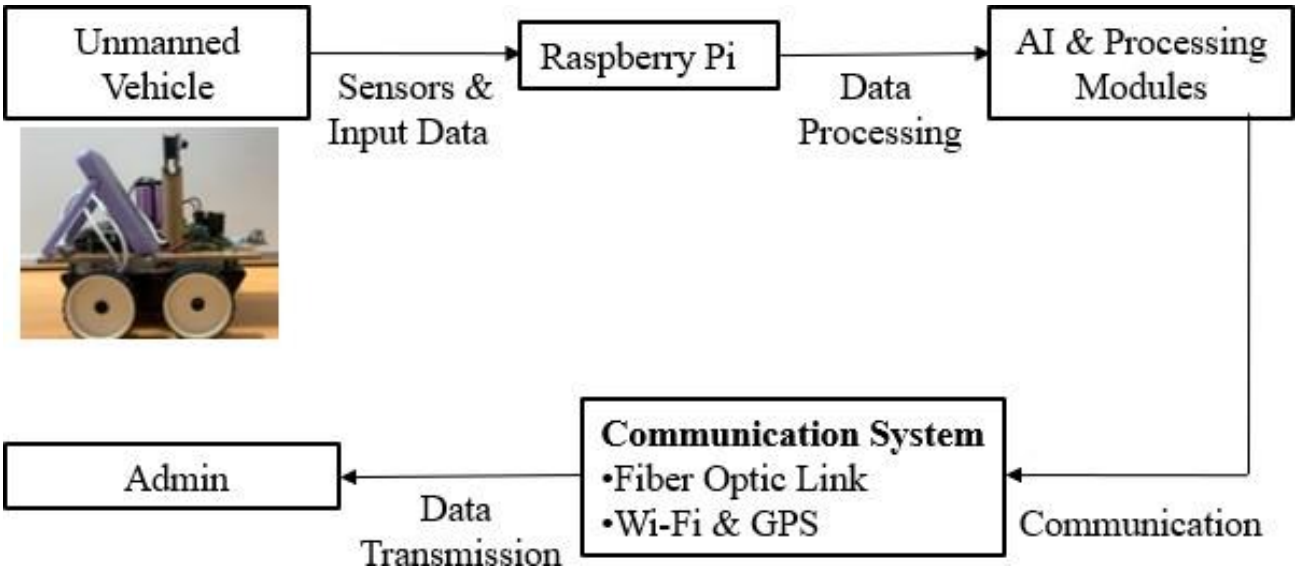

Figure 5.1 Proposed Architecture Diagram

The proposed system's AI-based learning models enable adaptability to various underground environments, improving performance over time. Machine learning algorithms analyze historical data and environmental patterns to refine navigation and hazard detection. The proposed system not only enhances scientific research but also minimizes the risks associated with traditional exploration methods, making cave studies more accessible and data-driven.

5.2 UML DIAGRAMS

5.2.1 Use Case Diagram

The diagram illustrates an environmental monitoring system that utilizes unmanned vehicles and IoT sensors for data collection and anomaly detection. The Admin is responsible for configuring IoT sensors and setting up unmanned vehicles, ensuring the system is ready for monitoring. The Operator plays a key role in monitoring the data, receiving alerts, and generating reports based on the findings. This system provides a structured approach to real-time monitoring, anomaly detection, and reporting, enhancing environmental analysis and decision-making.

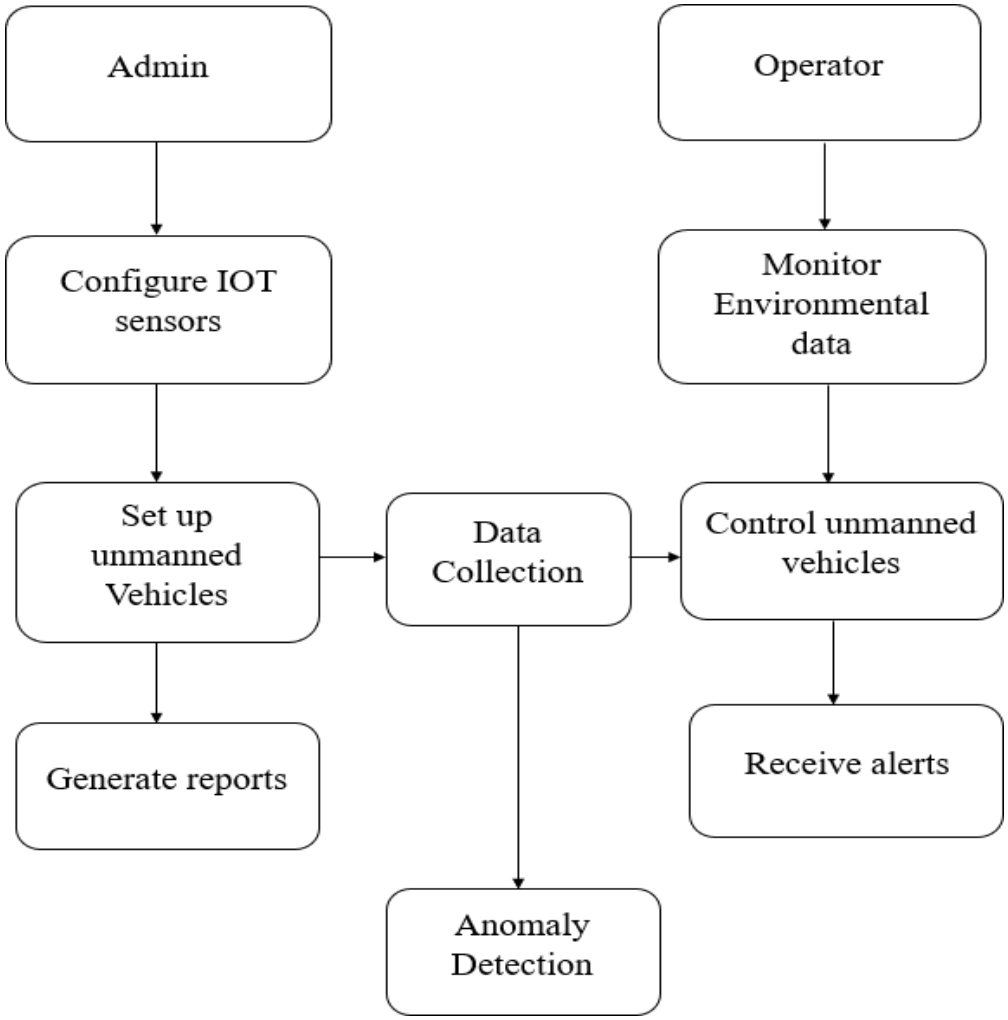

Figure 5.2.1 Use Case Diagram

5.2.2 Sequence Diagram

The sequence diagram illustrates the workflow of an Unmanned Vehicle Monitoring System, detailing the interactions between the User, Vehicle, Sensor, GPS, and Cloud Server. The process begins when the User initiates the system by sending a Start command to the Vehicle, which then activates its Sensors to collect environmental data. This data is processed and supplemented with GPS location information, retrieved from the GPS module and sent to the Cloud Server. The Cloud Server analyzes the data to detect any potential hazards. If a hazard is identified, an alert is generated and sent back to the Vehicle, which subsequently notifies the User. This real-time system enhances situational awareness and aids in decision-making by providing timely alerts.

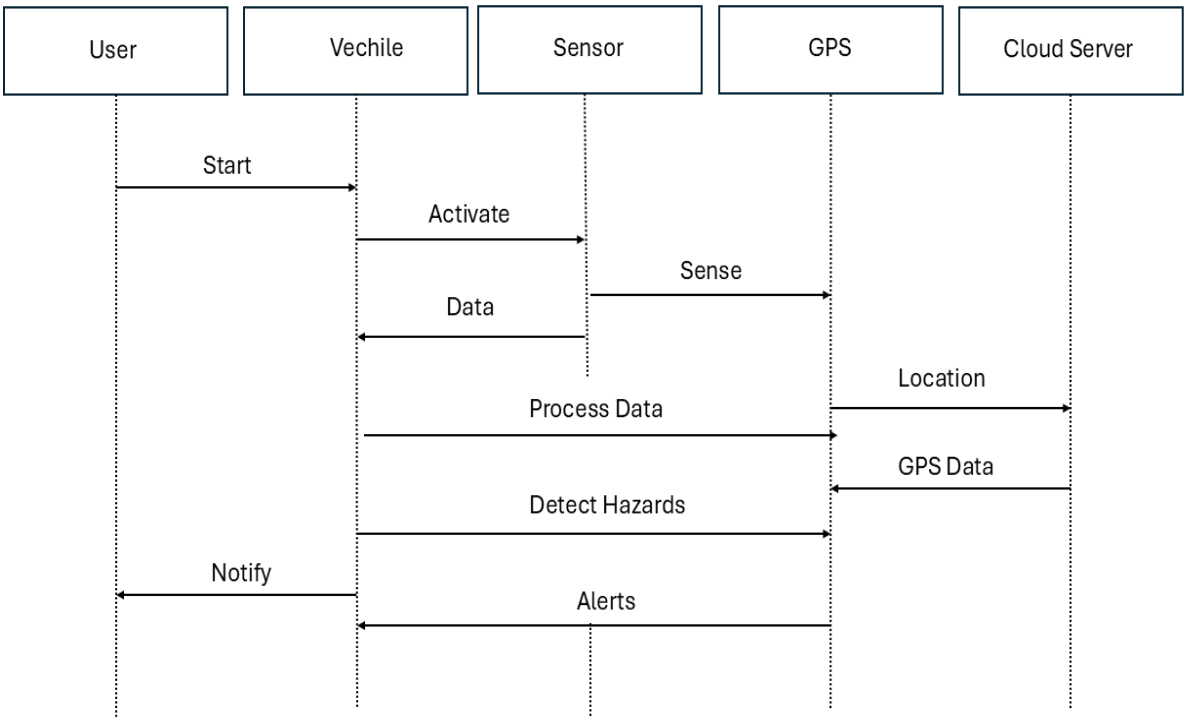

Figure 5.2.2 Sequence Diagram

5.3 DATA FLOW DIAGRAM

5.3.1 LEVEL 0

Provides a simplified view of the system, starting with sensors that collect environmental data like air quality and temperature. This data is processed by the unmanned vehicle monitor, which generates alerts. Finally, processed information is shared with users for remote monitoring and analysis of cave conditions.

Figure 5.3.1 System Overview

5.3.2 LEVEL 1

Breaks the system into major components: Air Quality Sensors detect toxic gases, Data Processing & AI Analysis identifies hazards, Navigation & Obstacle Avoidance ensures safe movement, and GPS Tracking provides real-time vehicle positioning. These components work together to create an autonomous and hazard-detecting cave exploration system.

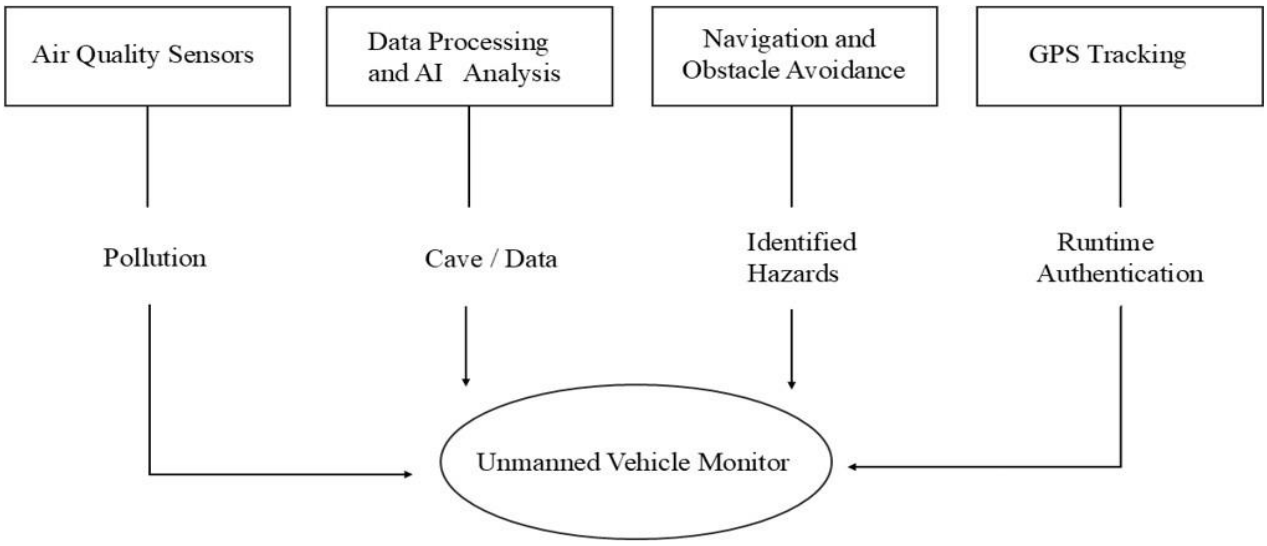

Figure 5.3.2 Functional Architecture

5.3.3 LEVEL 2

At the sensor module collects raw data, which is processed by the AI module to detect hazards. Based on this analysis, the navigation system adjusts the vehicle's movement, avoiding obstacles. The web dashboard displays real-time updates, alerts, and status information for researchers to monitor and act upon.

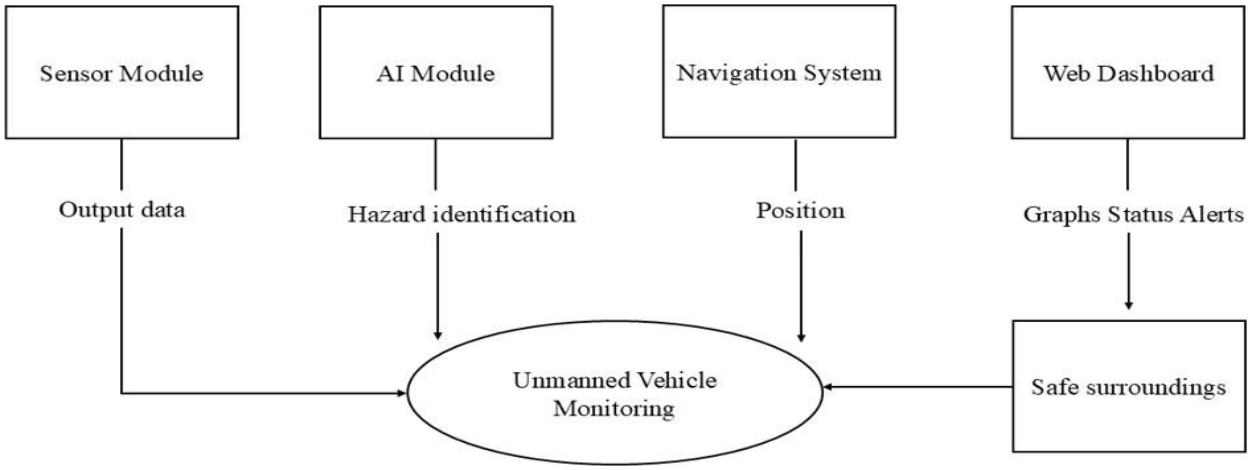

Figure 5.3.3 Module Interaction

5.3.4 LEVEL 3

It delves into detailed processing: the sensor module collects environmental data, which is analyzed by the AI to identify hazards. The navigation system adjusts vehicle movement accordingly, avoiding dangers. The processed data is then displayed on the user interface through visual dashboards, providing real-time insights for monitoring cave conditions. The AI continually refines its hazard detection algorithms, improving the system's accuracy over time.

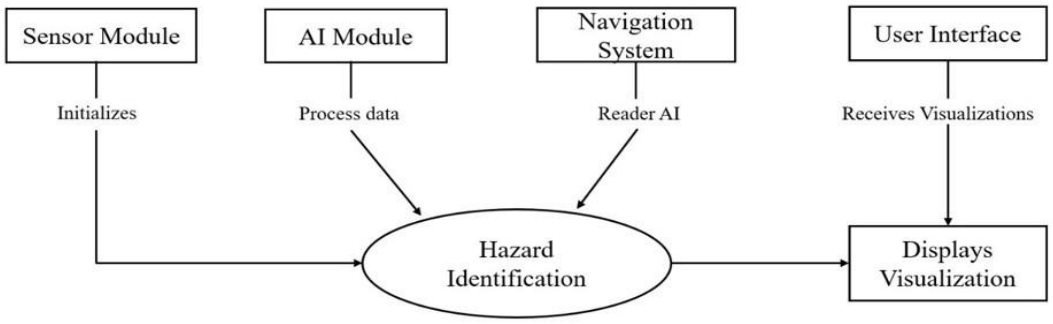

Figure 5.3.4 Processing And Data Flow

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 MODULES

- Environmental Sensing and Data Collection
- Autonomous Navigation and Obstacle Avoidance
- GPS Module
- User Interface and Data Visualization

6.2 MODULES DESCRIPTION

6.2.1 Environmental Sensing And Data Collection

This module integrates advanced sensors into an unmanned vehicle to collect critical environmental data. It continuously monitors air quality, temperature, humidity, lighting conditions, soil health, and obstacles. AI-driven algorithms process this real-time data to identify hazards such as toxic gases, unstable terrain, and sudden water inflows. By ensuring accurate and timely data collection, the system enhances safety for researchers and provides valuable insights for geologists, biologists, and other scientists studying cave ecosystems. The module's ability to detect environmental changes early helps prevent accidents and facilitates informed decision-making in challenging underground environments. Furthermore, the integration of AI enables automated hazard detection, reducing human intervention and improving efficiency. This technology is particularly useful for studying fragile ecosystems where human presence may be limited or unsafe.

6.2.2 Autonomous Navigation And Obstacle Avoidance

The unmanned vehicle features AI-based navigation and obstacle detection systems, enabling safe traversal through complex cave environments. Equipped with depth sensors and image processing techniques like the YOLO algorithm, it identifies obstacles in real time. The vehicle autonomously adjusts its path to avoid hazards such as rocks, water, and narrow passages, ensuring smooth navigation. An

autonomous retreat mechanism enhances safety by detecting dangerous conditions, such as high pollution levels or flooding. If such risks are identified, the vehicle reverses course to prevent damage and ensure mission continuity. This intelligent system minimizes human intervention, allowing researchers to explore challenging cave ecosystems more efficiently. By integrating AI-driven decision-making with real-time hazard detection, the vehicle enhances operational reliability. It provides valuable environmental data while ensuring self-preservation in extreme conditions. This technology supports geologists, biologists, and environmental scientists in conducting safer and more effective research in hazardous subterranean environments. Furthermore, the integration of AI enables automated hazard detection, reducing human intervention and improving efficiency. This technology is particularly useful for studying fragile ecosystems where human presence may be limited or unsafe.

6.2.1 GPS Module

The unmanned vehicle is equipped with a GPS tracking system that provides real-time location data, helping researchers map underground pathways and monitor its movement. By continuously transmitting precise coordinates, the system ensures accurate navigation through complex cave terrains. The GPS module plays a crucial role in preventing the vehicle from getting lost in deep or uncharted areas. It assists in retrieval operations by providing exact position updates, reducing the risk of losing valuable data and equipment. Additionally, the collected geospatial data supports scientific research, enabling geologists and biologists to study underground structures, ecosystems, and environmental changes over time. By integrating GPS tracking with advanced navigation technologies, the system enhances the efficiency and safety of cave exploration. It allows researchers to conduct studies with greater accuracy while minimizing risks associated with navigating challenging subterranean environments. This makes it a valuable tool for scientific discovery and exploration.

6.2.1 User Interface And Data Visualization

The web-based dashboard is an essential part of the system, allowing researchers and operators to monitor real-time data, visualize trends, and remotely control the vehicle. It features an intuitive interface displaying live environmental data, such as air quality, temperature, humidity, water levels, and lighting conditions. To enhance safety, the system generates automatic hazard alerts, notifying users of risks like pollution, flooding, or low visibility. This enables quick decision-making and timely responses to potential dangers. Advanced data visualization tools, including interactive maps and historical trend charts, help researchers analyze cave conditions over time. By studying past data, they can identify patterns, predict environmental changes, and improve exploration strategies.

6.3 ALGORITHM TECHNIQUES

1. IoT-Based Real-Time Data Collection

The IoT framework plays a crucial role in collecting real-time environmental data, including air quality, temperature, humidity, lighting, and water levels. Utilizing Raspberry Pi and various sensors, the system continuously monitors cave conditions, ensuring accurate data collection for informed decision-making and hazard detection. This enables researchers to assess environmental changes and potential risks efficiently. To enhance navigation, the system integrates the YOLO (You Only Look Once) algorithm for real-time object detection. YOLO processes visual data rapidly, allowing the unmanned vehicle to identify obstacles such as rocks, narrow passages, or water bodies with high precision. This AI-powered detection system enables smooth and safe navigation through complex cave environments. By combining IoT-based environmental monitoring with advanced AI-driven object detection, the system ensures enhanced safety, efficiency, and reliability in cave exploration. This technology supports scientists and researchers in conducting detailed underground studies with minimal human intervention.

2. Dehaze Algorithm for Improved Visibility

Low-light and foggy conditions in caves can make it difficult to detect obstacles and capture clear images. To solve this problem, the system uses a dehaze algorithm that improves visibility by reducing noise and enhancing image contrast. This helps in getting clearer images, which improves navigation and analysis.The AI-based hazard detection and decision-making system allows the unmanned vehicle to analyze its surroundings and take necessary actions. By processing data from environmental sensors and image inputs, the AI can detect dangers like air pollution, flooding, and blocked pathways. If the system identifies any risks, it can change the vehicle's path or retreat to avoid danger.This combination of image enhancement and AI-driven hazard detection helps the vehicle move safely through caves. It improves overall efficiency, ensuring that researchers can collect accurate data while minimizing risks in challenging underground environments.

3. AI-Based Decision-Making for Hazard Detection

The system utilizes AI-driven decision-making to detect hazards such as air pollution and flooding, allowing the vehicle to operate autonomously without human intervention. This improves efficiency in cave exploration by enabling real-time analysis of environmental conditions and immediate response to potential dangers.To further enhance safety, the vehicle is equipped with an autonomous retreat mechanism. If an obstacle blocks the path or environmental conditions become too dangerous, the system automatically calculates a safe route and returns the vehicle to a secure location. This prevents it from getting trapped or damaged in hazardous areas.By combining AI-based hazard detection with an intelligent retreat mechanism, the system ensures reliable and efficient cave navigation. It minimizes risks, improves data collection accuracy, and supports safe exploration for researchers. This technology plays a crucial role in studying underground ecosystems while reducing the need for direct human involvement in dangerous environments.

4. Autonomous Retreat Mechanism

To prevent accidents and system failures, the vehicle is designed with an autonomous retreat mechanism. If hazardous conditions such as high pollution levels, flooding, or unstable terrain are detected, the system automatically triggers a retreat, ensuring the vehicle moves to a safer location. This feature helps protect both the vehicle and the data it collects.

For effective navigation, the system uses an image-processing algorithm that divides images into grids and predicts bounding boxes with class probabilities. This allows the vehicle to accurately detect and recognize obstacles such as rocks, stalactites, narrow paths, and water bodies. With this information, the vehicle can adjust its path to avoid collisions and ensure smooth movement through the cave.

By combining real-time hazard detection with AI-driven obstacle recognition, the system enhances safety and reliability. This makes it a valuable tool for researchers studying underground environments, ensuring efficient exploration while minimizing risks.

5. GPS Tracking for Navigation and Research

The system uses GPS tracking to provide accurate location data, allowing researchers to monitor the vehicle's movement in real time. This helps in mapping underground pathways, ensuring smooth navigation, and preventing the vehicle from getting lost in complex cave structures. With precise tracking, researchers can analyze the vehicle's path, improve exploration strategies, and retrieve it easily if needed. The GPS data also supports scientific studies by providing detailed maps of cave environments. By integrating GPS tracking with advanced technologies like AI-driven navigation and real-time hazard detection, the system ensures a reliable and efficient approach to cave exploration. It improves data accuracy, enhances safety, and reduces risks for researchers working in challenging underground conditions. This combination of features makes the system a valuable tool for geologists, biologists, and environmental scientists studying cave ecosystems.

CHAPTER 7

SYSTEM TESTING

7.1 HARDWARE TESTING

To ensure the reliability and efficiency of the underground monitoring system, a comprehensive range of hardware testing techniques is implemented. These testing methods are essential for validating the overall functionality, compatibility, durability, and performance of the unmanned vehicle and its integrated components. The goal is to ensure that the system can operate reliably in the unpredictable and challenging conditions of cave exploration.Functional testing is conducted to verify that each hardware component, such as sensors, actuators, and communication modules, performs as intended. For instance, the air quality sensor (MQ135) is tested in controlled environments to confirm its accuracy in detecting gas concentrations. Similarly, temperature and humidity sensors are evaluated to ensure reliable data collection.

Compatibility testing ensures seamless interaction between various system components. The Raspberry Pi microcontroller is tested to verify that it can efficiently process data from multiple sensors without interference or data loss. Communication between sensors, actuators, and navigation systems is also assessed for smooth operation. Environmental testing simulates real-world conditions such as low visibility, high humidity, and uneven terrain to evaluate system durability. The ESP32 camera is tested in dark and damp environments to determine its effectiveness in capturing clear images. Power consumption testing is also performed to optimize energy efficiency and extend mission duration. By integrating these testing techniques, the system is fine-tuned for maximum reliability. These rigorous evaluations help ensure that all hardware components work effectively both independently and as part of the system’s integrated operations, providing accurate data and supporting safe, efficient cave exploration.

7.2 TESTING TECHNIQUES

1.Functional Testing

Functional testing is a critical step in ensuring that each hardware component in the underground monitoring system operates as intended. The primary objective is to validate the functionality and performance of individual parts, including sensors, actuators, and other integrated modules. For instance, the air quality sensor (MQ135) is tested in controlled environments with different gas concentrations. Its readings are then compared to known reference values to confirm accuracy and consistency. Similarly, temperature and humidity sensors are exposed to varying conditions to ensure reliable data collection. Actuators responsible for vehicle movement and obstacle avoidance are also tested to verify smooth operation and responsiveness. By conducting functional testing, engineers can detect faulty components, correct calibration errors, and optimize system performance. This ensures that all hardware elements work efficiently under real-world conditions, improving the reliability of underground exploration. Proper functional validation helps guarantee that the system provides accurate environmental data, enabling researchers to make informed decisions and conduct successful cave studies.

2.Compatibility Testing

Compatibility testing is essential to ensure that all hardware components of the unmanned vehicle work seamlessly together. This process validates the communication and interaction between sensors, actuators, microcontrollers, and other integrated modules, ensuring smooth data flow and system functionality. A key aspect of this testing involves verifying that the Raspberry Pi microcontroller can efficiently collect and process data from multiple sensors without interference or data loss. Sensors measuring temperature, humidity, air quality, and soil conditions are tested to confirm that they transmit accurate readings while operating simultaneously. Proper functional validation helps guarantee that the

system provides accurate environmental data, enabling researchers to make informed decisions and conduct successful cave studies. Additionally, actuators responsible for movement and obstacle avoidance are assessed to ensure proper coordination with the control system. By conducting compatibility testing, engineers can detect and resolve integration issues, optimize data processing efficiency, and prevent communication failures. This helps improve the system's reliability and performance in real-world applications. Ensuring seamless hardware compatibility is crucial for the vehicle's successful operation, allowing it to function effectively during cave exploration missions while providing accurate environmental data.

3.Environmental Testing

Environmental testing is crucial to evaluating the unmanned vehicle's performance in real-world, challenging conditions. Since the vehicle is designed for cave exploration, it must function effectively in environments with low visibility, high humidity, and uneven terrain. This testing ensures that all sensors and systems operate reliably despite these harsh conditions. Key aspects of testing include assessing how well the vehicle navigates rocky or slippery surfaces, maintains communication in enclosed spaces, and withstands exposure to moisture. The ESP32 camera, for example, is tested in dark and damp environments to determine its ability to capture clear images under poor lighting conditions. Similarly, environmental sensors are evaluated for accuracy in detecting temperature, humidity, air quality, and water levels. By conducting rigorous environmental testing, engineers can identify potential failures and improve the vehicle's durability. This ensures it remains operational during long exploration missions, providing researchers with accurate and reliable data while minimizing the risk of system malfunctions in extreme cave conditions.

4.Power Consumption Testing

Power consumption testing is essential to ensure the unmanned vehicle operates efficiently during extended missions. Since the vehicle depends on battery power, analyzing its energy usage under various operational conditions helps optimize performance and extend mission duration. The testing process involves measuring power draw when the vehicle performs key functions such as collecting sensor data, navigating obstacles, and transmitting information in real time. By monitoring energy usage in different scenarios, engineers can identify components that consume the most power and implement strategies to improve efficiency. Additionally, testing helps determine the optimal battery capacity required for long-duration missions. Engineers analyze power consumption trends to refine energy management strategies, such as optimizing sensor usage, adjusting processing loads, and improving battery efficiency. This helps improve the system's reliability and performance in real-world applications. Ensuring seamless hardware compatibility is crucial for the vehicle's successful operation, allowing it to function effectively during cave exploration missions while providing accurate environmental data. By conducting thorough power consumption testing, the system can be fine-tuned to maximize battery life, reduce downtime, and ensure continuous operation in challenging environments. This enhances the overall reliability of the vehicle, making it a more effective tool for cave exploration and scientific research.

5.Reliability Testing

Reliability and stress testing are essential to assess the long-term durability and stability of the unmanned vehicle's hardware. These tests ensure that the components can withstand continuous operation and harsh environmental stressors such as heat, humidity, and mechanical wear. During stress testing, the system is subjected to prolonged operational periods, simulating real-world cave exploration conditions. The vehicle is exposed to extreme temperatures, moisture, and rugged

terrains to evaluate its resilience. The performance of sensors, motors, power supply, and communication modules is closely monitored to identify potential failures or weaknesses. By analyzing the data collected during these tests, engineers can fine-tune the system's design, improve component reliability, and enhance overall durability. These evaluations help ensure that the vehicle can function efficiently in challenging underground environments without frequent breakdowns. Ultimately, reliability and stress testing play a crucial role in optimizing the system for long-term use.

They enhance safety, improve mission success rates, and ensure that researchers can rely on the vehicle for accurate and uninterrupted data collection. The testing process involves measuring power draw when the vehicle performs key functions such as collecting sensor data, navigating obstacles, and transmitting information in real time. By monitoring energy usage in different scenarios, engineers can identify components that consume the most power and implement strategies to improve efficiency.

Additionally, testing helps determine the optimal battery capacity required for long-duration missions. Engineers analyze power consumption trends to refine energy management strategies, such as optimizing sensor usage, adjusting processing loads, and improving battery efficiency. By conducting compatibility testing, engineers can detect and resolve integration issues, optimize data processing efficiency, and prevent communication failures. This helps improve the system's reliability and performance in real-world applications. Ensuring seamless hardware compatibility is crucial for the vehicle's successful operation, allowing it to function effectively during cave exploration missions while providing accurate environmental data.

CHAPTER 8

EXPERIMENTAL RESULT AND DICUSSION

8.1 EXPERIMENTAL RESULTS

8.1.1 Data Collection Analysis

The autonomous cave exploration vehicle is built to operate effectively in harsh environments, incorporating advanced sensors for environmental monitoring, hazard detection, and real-time data transmission. It employs a DHT11 sensor to measure temperature and humidity, while air quality is assessed using MQ2, MQ8, and MQ135 sensors to detect hazardous gases such as methane and carbon monoxide. Additionally, a capacitive soil sensor monitors moisture levels to evaluate terrain stability.

Figure 8.1.1 System Requirements

8.1.2 System Deployment

The autonomous cave exploration vehicle is developed in multiple phases, integrating AI, IoT, robotics, and communication technologies to ensure efficient and dependable operation in extreme environments. The project is designed for real-time data collection, hazard detection, and autonomous navigation within caves. During the conceptualization and planning phase, key technologies are

selected, including YOLO for object detection, dehaze algorithms for image enhancement, fiber optic communication for high-speed data transfer, and AI-driven anomaly detection.

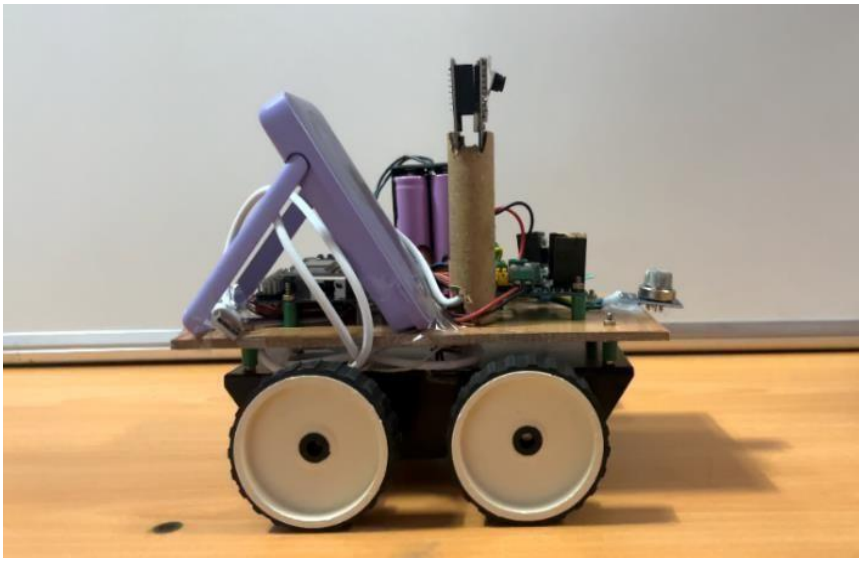

Figure 8.1.2 Deployment Model

Raspberry Pi microcontroller, a DHT11 sensor for temperature and humidity for monitoring, MQ sensors for detecting hazardous gases, and a capacitive soil sensor for evaluating terrain stability.

8.1.3 Result and Interpretation

The autonomous cave exploration vehicle gathered real-time environmental data, including temperature, humidity, air quality, and terrain stability, transmitting it to cloud storage for further analysis. An ESP32 camera, enhanced with dehaze algorithms, captured clear visuals even in low-light conditions. AI-powered YOLO enabled precise obstacle detection, while MQ sensors monitored hazardous gases. The vehicle autonomously navigated through confined spaces using AI, GPS, and fiber optic communication, ensuring accurate tracking. Extensive lab, simulation, and field testing validated its reliability. The integration of AI, IoT, and cloud storage significantly improved safety and efficiency, with future developments focusing on enhanced battery life and

advanced navigation. Raspberry Pi microcontroller, a DHT11 sensor for temperature and humidity for monitoring, MQ sensors for detecting hazardous gases, and a capacitive soil sensor for evaluating terrain stability.

8.2 SUMMARY OF DISCUSSION

The Real-Time Underground Monitoring System Using an Unmanned Vehicle improves cave exploration by integrating IoT, AI, Raspberry Pi, and GPS. It addresses risks such as low visibility, structural instability, and hazardous air conditions through sensor-based environmental monitoring and obstacle detection, transmitting real-time data remotely.Key innovations include AI-driven decision-making, GPS tracking, and real-time visualization, enhancing both safety and efficiency. The system uses YOLO for object detection, a dehaze algorithm for improved visibility, and AI-based navigation to optimize movement, benefiting scientific research and geological surveys.

Designed with advanced hardware and software, it features ultrasonic, air quality, temperature, humidity, and soil sensors, along with an ESP32 camera and GPS module. Its IoT-enabled software allows remote monitoring through a web or mobile dashboard.The system is structured into four main modules: Environmental Sensing, AI Navigation, GPS Tracking, and Data Visualization. It continuously monitors environmental conditions, avoids obstacles, logs locations, and provides real-time insights through an interactive interface.Challenges such as sensor precision, weak GPS signals, and harsh environmental conditions require solutions like inertial navigation and durable hardware. Future enhancements may include advanced AI, automation, and improved sensors, with potential applications in mining safety, disaster response, and archaeology. This AI-driven vehicle revolutionizes underground exploration by ensuring safer.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENT

9.1 CONCLUSION

In summary, this system represents a major breakthrough in cave exploration by utilizing unmanned vehicles equipped with IoT, Raspberry Pi, AI, and advanced sensors. The seamless integration of these technologies enables efficient, safe, and autonomous monitoring of cave environments, providing researchers with crucial real-time data while reducing human risk. The vehicle's capability to navigate challenging and hazardous cave conditions, detect potential dangers, and remotely transmit valuable insights marks significant progress in both exploration and environmental monitoring.Additionally, its modular design allows for customization based on different cave conditions and research requirements, making it a versatile tool for future exploration. Overall, this innovative approach enhances safety, data collection, and research, contributing to a better understanding of these vital natural ecosystems.

9.2 FUTURE ENHANCEMENT

In the future, the autonomous cave exploration vehicle could be upgraded with an efficient solar power system to enable continuous operation in remote, off-grid locations. This enhancement would minimize reliance on conventional battery charging, significantly extending the vehicle's operational lifespan during extended missions. The system would operate in two phases: solar energy integration and battery backup. Lightweight, high-efficiency solar panels would be installed to harness sunlight and charge the battery during daylight hours. When solar energy is insufficient, the system would seamlessly switch to battery power, ensuring uninterrupted functionality in dark or low-light conditions.

APPENDICES

A1 HARDWARE DESCRIPTION

1. Microcontrollers and Processing Units

Microcontrollers play a vital role in processing sensor data and controlling the vehicle's movements. The ESP32-CAM is used for capturing real-time images and video footage of underground conditions, enabling remote visual monitoring. The ESP32 Controller acts as the main processing unit, integrating all sensor data and managing communication. Additionally, the ESP8266 Controller is used for wireless data transmission, ensuring seamless connectivity between the vehicle and the monitoring system.

2. Sensors for Environmental Monitoring

The system incorporates various sensors to monitor underground conditions effectively. The DHT11 sensor is responsible for measuring temperature and humidity, ensuring that underground conditions remain within safe limits. A soil moisture sensor is integrated to detect moisture levels, which is crucial for soil stability analysis in mining and construction applications. Gas detection is handled by the MQ-4 sensor, which detects methane (CH4) to prevent the risk of explosions, and the MQ-135 sensor.

3. Communication System

For real-time data transmission, the system is equipped with Wi-Fi modules (ESP8266/ESP32), enabling wireless communication and data transfer to a cloud server or a remote monitoring center. In areas with limited connectivity, an optional LoRa module can be used for long-range data transmission. These communication modules ensure continuous and reliable data flow, allowing users to monitor underground conditions remotely.

4. Power Supply

A stable power source is essential for the continuous operation of the unmanned vehicle and its sensors. The system is powered by a Lithium-Ion battery (7.4V - 12V, 2500mAh or higher), ensuring uninterrupted operation. To regulate the voltage supplied to different components, a voltage regulator (AMS1117 3.3V/5V) is used, preventing damage due to power fluctuations. In remote underground locations, a solar panel can be integrated as an optional power source, ensuring long-term sustainability and reduced dependency on battery charging.

5.Additional Components

To enable vehicle movement, motor drivers (L298N or L293D) are used to control the DC motors, allowing precise navigation in underground environments. Infrared (IR) sensors can be integrated for obstacle detection, helping the vehicle navigate autonomously without collisions. Additionally, an OLED display can be included to provide real-time sensor readings directly on the vehicle, improving on-site monitoring capabilities.

A2 SOFTWARE DESCRIPTION

1. Server (Backend System)

The server is responsible for handling communication between the sensors, database, and client interface. It is developed using Node.js with Express.js or Django (Python) to manage data requests efficiently. The ESP32 and ESP8266 microcontrollers send sensor data to the server using HTTP POST requests or MQTT protocol. The server processes incoming data, applies filtering algorithms to remove noise, and forwards it to the database for storage. Additionally, the server implements real-time communication using Web Sockets to ensure live data updates on the client side without delays. Security measures, such as JWT authentication and SSL encryption, are added to protect data integrity and prevent unauthorized access.

2. Client (Web-Based Dashboard)

The client-side is designed as a responsive web application where users can monitor underground environmental conditions in real time. The front-end is built using HTML, CSS, and JavaScript, with frameworks like React.js or Vue.js for an interactive dashboard. The client retrieves sensor readings from the server via RESTful APIs or WebSockets, ensuring instant updates without refreshing the page. Data visualization is implemented using Chart.js or D3.js, displaying temperature, humidity, soil moisture, and gas levels in graphical formats. The interface also includes alert notifications, triggering warnings when sensor readings exceed safety thresholds. Additionally, the client supports multi-device access, allowing users to monitor underground conditions from desktops, tablets, and smartphones.

3. Database (Data Storage and Management)

The database is responsible for storing sensor readings and historical logs for analysis. A cloud-based database such as Firebase, MySQL, or MongoDB is used to handle large volumes of sensor data efficiently. The database schema is designed to store sensor ID, timestamp, and corresponding readings for temperature, humidity, gas levels, and soil moisture.

Data retrieval is optimized for quick access, enabling real-time updates on the client side. Additionally, backup mechanisms and cloud storage are implemented to ensure data availability in case of system failures. By integrating a powerful backend server, an interactive client-side dashboard, and a scalable database, this software ensures real-time monitoring, efficient data management, and a seamless user experience for underground environmental monitoring.

A3 SAMPLE CODE

```
#include <AsyncTCP.h>
#include <ESPAsyncWebServer.h>
#include <WebSocketsServer.h>
#include <DHT.h>
// WiFi Credentials
const char* ssid = "project";
const char* password = "1234567890";
// Motor Driver Pins (L298N)
#define IN1 26
#define IN2 27
#define IN3 14
#define IN4 12
// Sensor Pins
#define DHTPIN 4
#define DHTTYPE DHT11
#define MQ135_PIN 34
#define MQ4_PIN 35
#define SOIL_MOISTURE_PIN 32 // Added Soil Moisture Sensor
DHT dht(DHTPIN, DHTTYPE);
AsyncWebServer server(80);
WebSocketsServer webSocket(81);
// Function Prototypes
void webSocketEvent(uint8_t num, WStype_t type, uint8_t *payload, size_t length);
String getSensorData();
void moveForward();
void moveBackward();
void turnLeft();
void turnRight();
```

```
void stopMotors();
// HTML Webpage
const char webpage[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
    <title>ESP32 Rover Control</title>
    <style>
        body { font-family: Arial, sans-serif; text-align: center; }
        .btn { padding: 10px; margin: 5px; font-size: 18px; cursor: pointer; }
        #data { font-size: 20px; margin-top: 20px; }
        #status { font-size: 24px; margin-top: 30px; font-weight: bold; color: green; }
    </style>
</head>
<body>
    <h2>ESP32 Rover Control</h2>
    <button class="btn" onclick="sendCommand('forward')">Forward</button><br>
    <button class="btn" onclick="sendCommand('left')">Left</button>
    <button class="btn" onclick="sendCommand('stop')">Stop</button>
    <button class="btn" onclick="sendCommand('right')">Right</button><br>
    <button class="btn" onclick="sendCommand('backward')">Backward</button>
    <h3>Sensor Data</h3>
    <div id="data">Loading...</div>
    <div id="status">Safe Surrounding</div>
    <script>
        var ws = new WebSocket("ws://" + window.location.hostname + ":81/");
        function sendCommand(cmd) {
            ws.send(cmd);
            console.log("Command Sent: " + cmd);
```

```
    }
    function fetchSensorData() {
        fetch('/sensor')
        .then(response => response.json())
        .then(data => {
            document.getElementById("data").innerHTML =
                "Temperature: " + data.temperature + "°C<br>" +
                "Humidity: " + data.humidity + "%<br>" +
                "MQ135: " + data.mq135 + "<br>" +
                "MQ4: " + data.mq4 + "<br>" +
                "Soil Moisture: " + data.soilMoisture;
        });
    }
// Motor Control Pins as Output
pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);
pinMode(IN3, OUTPUT);
pinMode(IN4, OUTPUT);
pinMode(SOIL_MOISTURE_PIN, INPUT); // Soil Moisture Sensor as Input
WiFi.begin(ssid, password);
Serial.print("Connecting to WiFi");
while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.print(".");
}
server.on("/", HTTP_GET, [](AsyncWebServerRequest *request) {
    request->send_P(200, "text/html", webpage);
});
server.on("/sensor", HTTP_GET, [](AsyncWebServerRequest *request) {
```

```
        String json = getSensorData();
        request->send(200, "application/json", json);
    });
    server.begin();
}
void loop() {
    webSocket.loop();
}
// WebSocket Event Handler
void webSocketEvent(uint8_t num, WStype_t type, uint8_t *payload, size_t length) {
    String command = String((char*)payload);
    command.trim();
    Serial.print("Command Received: ");
    Serial.println(command);
    if (command == "forward") {
        moveForward();
    } else if (command == "backward") {
        moveBackward();
    } else if (command == "left") {
        turnLeft();
    } else if (command == "right") {
        turnRight();
    } else if (command == "stop") {
        stopMotors();
    }
}
void turnLeft() {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
```

```
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
}
void turnRight() {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
}
void stopMotors() {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
}
String getSensorData() {
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();
    int mq135 = analogRead(MQ135_PIN);
    int mq4 = analogRead(MQ4_PIN);
    int soilMoisture = analogRead(SOIL_MOISTURE_PIN);
    String json = "{";
    json += "\"temperature\":" + String(temperature) + ",";
    json += "\"humidity\":" + String(humidity) + ",";
    json += "\"mq135\":" + String(mq135) + ",";
    json += "\"mq4\":" + String(mq4) + ",";
    json += "\"soilMoisture\":" + String(soilMoisture);
    json += "}";
    return json;}
```

```
#include "WiFi.h"
#include "app_httpd.h"
#include "camera_pins.h"

// WiFi Credentials
const char *ssid = "your_SSID";
const char *password = "your_PASSWORD";

void setup() {
  Serial.begin(115200);
  Serial.println("\nStarting ESP32-CAM...");

  // Camera Config
  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer = LEDC_TIMER_0;
  config.pin_d0 = Y2_GPIO_NUM;
  config.pin_d1 = Y3_GPIO_NUM;
  config.pin_d2 = Y4_GPIO_NUM;
  config.pin_d3 = Y5_GPIO_NUM;
  config.pin_d4 = Y6_GPIO_NUM;
  config.pin_xclk = XCLK_GPIO_NUM;
  config.pin_pclk = PCLK_GPIO_NUM;
  config.pin_vsync = VSYNC_GPIO_NUM;
  config.pin_href = HREF_GPIO_NUM;
  config.pin_sccb_sda = SIOD_GPIO_NUM;
  config.pin_sccb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn = PWDN_GPIO_NUM;
  config.pin_reset = RESET_GPIO_NUM;
```

```
    config.xclk_freq_hz = 20000000;
    config.pixel_format = PIXFORMAT_JPEG;
    config.frame_size = FRAMESIZE_UXGA;
    config.fb_count = 1;
    if (psramFound()) {
        config.jpeg_quality = 10;
        config.fb_count = 2;
    } else {
        config.frame_size = FRAMESIZE_SVGA;
        config.fb_location = CAMERA_FB_IN_DRAM;
    }
    if (esp_camera_init(&config) != ESP_OK) {
        Serial.println("Camera init failed!");
        return;
    }
    WiFi.begin(ssid, password);
    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
    Serial.println("\nWiFi Connected!");
    Serial.print("Visit: http://");
    Serial.println(WiFi.localIP());
    startCameraServer();
}
void loop() {
    delay(10000);
}
```

```
void stopMotors() {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
}

String getSensorData() {
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();
    int mq135 = analogRead(MQ135_PIN);
    int mq4 = analogRead(MQ4_PIN);
    int soilMoisture = analogRead(SOIL_MOISTURE_PIN);

    String json = "{";
    json += "\"temperature\":" + String(temperature) + ",";
    json += "\"humidity\":" + String(humidity) + ",";
    json += "\"mq135\":" + String(mq135) + ",";
    json += "\"mq4\":" + String(mq4) + ",";
    json += "\"soilMoisture\":" + String(soilMoisture);
    json += "}";

    return json;
}
```

A4. SCREENSHOTS

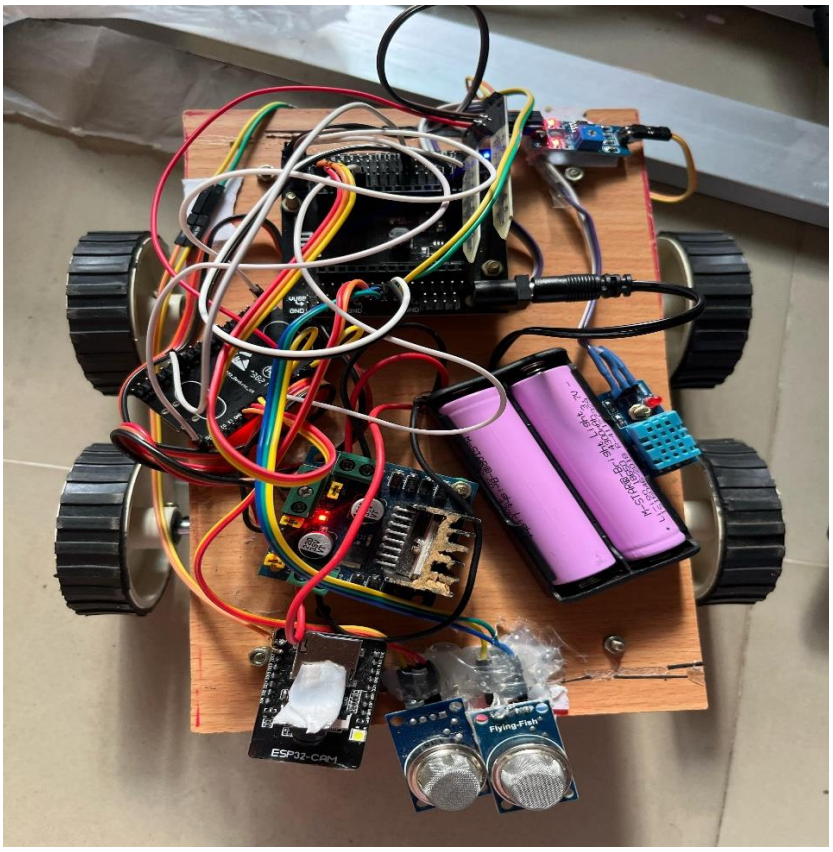

FIGURE A4.1 Unmanned Vehicle

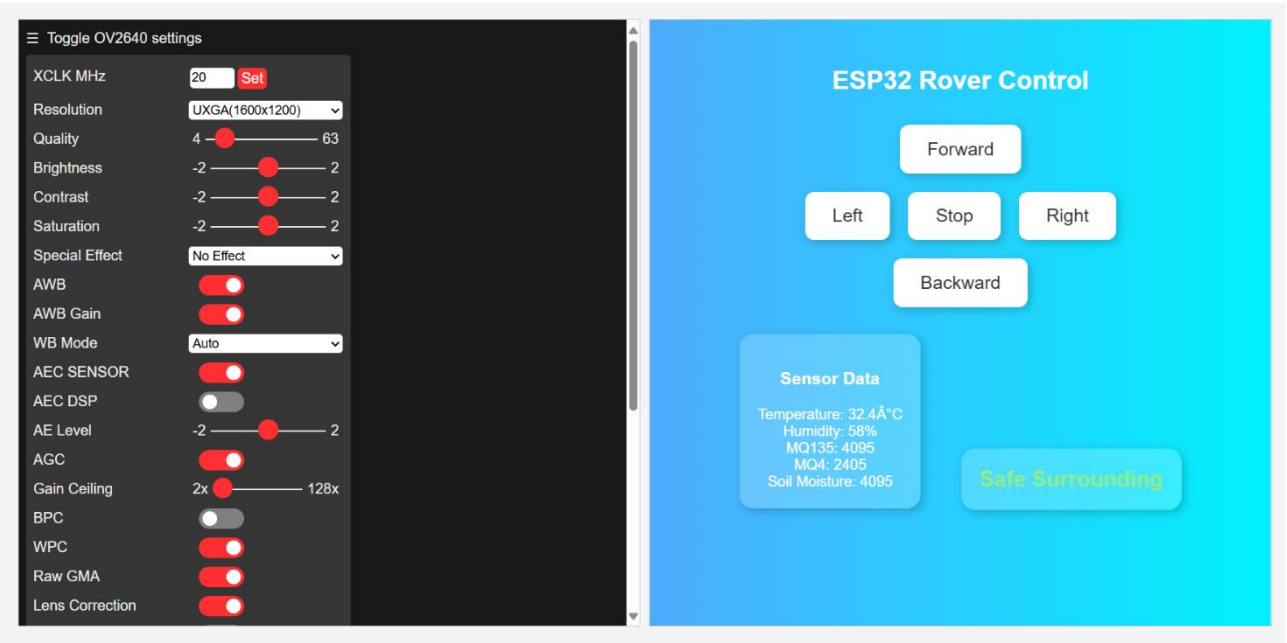

FIGURE A4.2 Control Website

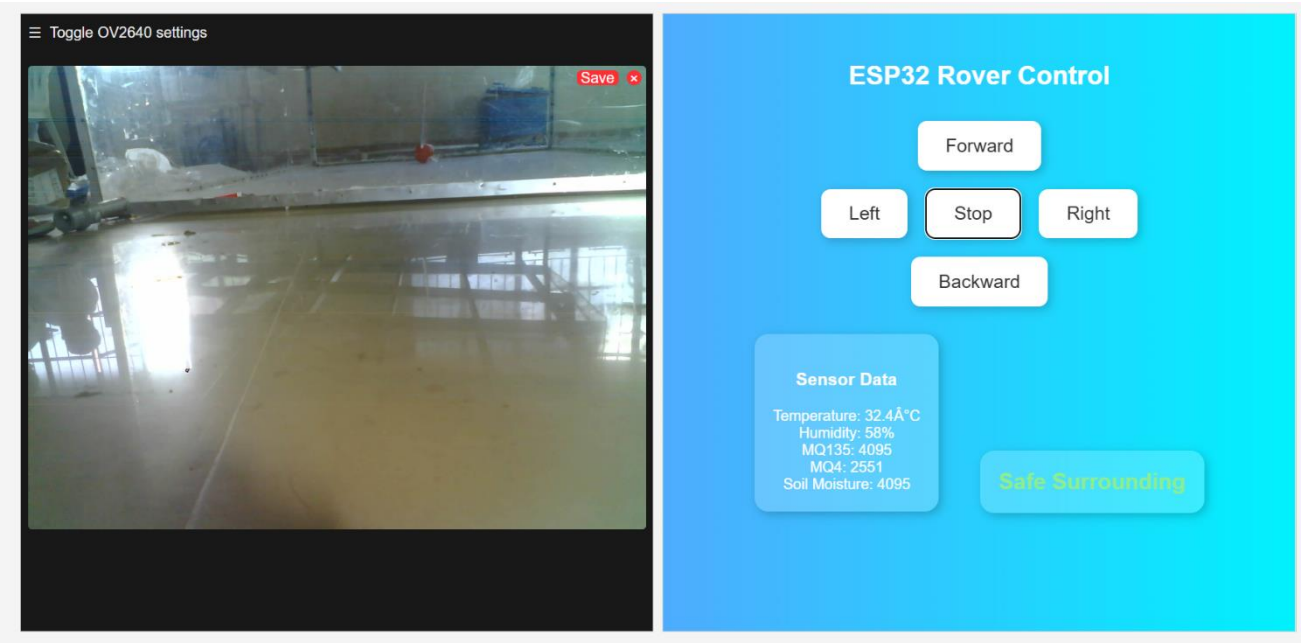

FIGURE A4.3 Video Monitoring

FIGURE A4.4 Original Image Into Enhanced Image

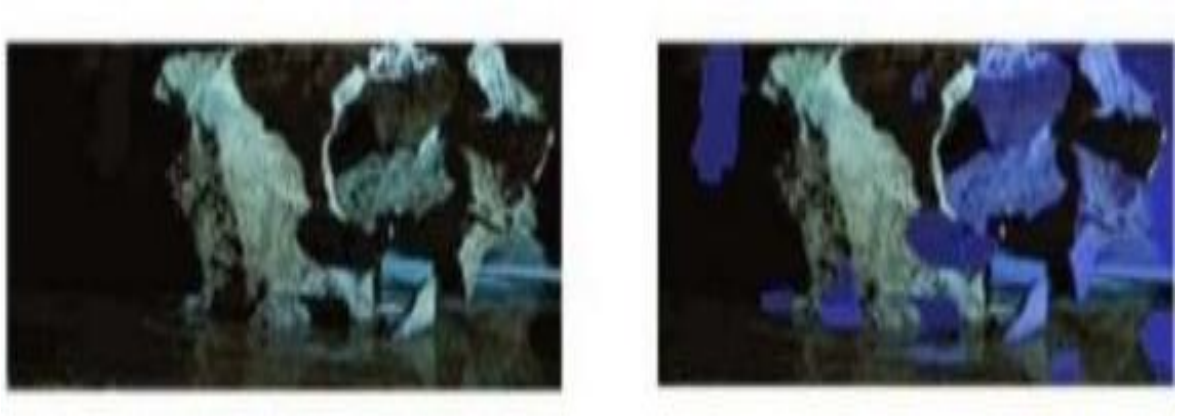

FIGURE A4.5 Enhanced Image Into Obstacles Detection

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LIST OF PUBLICATIONS

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